



Strings Problem Bank (Java)

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to List (ArrayList & LinkedList).

Section 1: String Basics and Creation

★ Create and Display a String

★ String Concatenation using + Operator

String Concatenation using concat()

★ Access Characters using charAt()

★ Find Length of a String

★ String Immutability Demonstration

Creating Strings using new String() vs Literal

Section 2: String Comparison (Java-Specific)

★ Compare Strings using equals()

★ Compare Strings using == Operator (Reference Pitfall)

★ Compare Strings Ignoring Case using equalsIgnoreCase()

★ Compare Strings Lexicographically using compareTo()

String Pool Behavior Demonstration

Check if Two Strings Point to the Same Reference

Section 3: String Methods and Manipulation

★ Extract Substring using substring()

★ Find Index of a Character or Substring using indexOf()

★ Replace Characters or Substrings using replace()

★ Trim Whitespace using trim() / strip()

★ Split a String using split()

Convert String to Character Array using toCharArray()

Convert Character Array Back to String

Check if String Contains a Substring

Check if String Starts or Ends With a Given Substring

Section 4: StringBuilder and StringBuffer (Java-Specific)

★ Build a String using StringBuilder



★ **Reverse a String using StringBuilder**

★ **Append, Insert, and Delete using StringBuilder**

StringBuilder vs StringBuffer vs String Performance Comparison

★ **Convert StringBuilder to String**

Section 5: Case Conversion and Formatting

★ **Convert String to Uppercase / Lowercase**

★ **Toggle Case of Each Character**

Capitalize First Letter of Each Word

★ **Format Strings using String.format()**

Pad a String with Leading or Trailing Characters

Section 6: String Searching and Pattern Matching

★ **Count Occurrences of a Character**

Count Occurrences of a Word

Find All Indexes of a Substring

★ **Check if String Matches a Pattern using matches() (Regex Basics)**

★ **Validate Email Format using Regex**

Validate Phone Number using Regex

Section 7: String-Based Logical Problems

★ **Check if a String is a Palindrome**

★ **Reverse a String without StringBuilder**

★ **Check if Two Strings are Anagrams**

★ **Count Vowels and Consonants in a String**

Remove Duplicate Characters from a String

★ **Find the First Non-Repeated Character**

Find the Most Frequent Character

Check if a String has All Unique Characters

Remove a Given Character from a String

★ **Word Frequency Counter in a Sentence**

Section 8: String Parsing and Tokenizing

★ **Split a Sentence into Words**

Count Number of Words in a Sentence

Reverse the Order of Words in a Sentence

★ **Reverse Each Word in a Sentence**



Problem Statements

Convert a CSV Line into an Array of Values

Parse and Sum Numbers Embedded in a String

Strings (Java)

Section 9: Real World String Applications

★ Password Strength Validator

★ Username Validator

Sentence Case Formatter

★ Palindrome Sentence Checker (Ignoring Spaces/Case)

Text Alignment / Justification Simulator

★ Log Message Parser